Assignment 3 -Risk Log & HeatMap

ALY6130 -Risk Management Analytics

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Introduction

This report documents the risk assessment conducted for EQB Inc.'s announced acquisition of PC Financial from Loblaw Companies Limited (December 2025). The deal transfers approximately 3.5 million customers and a credit card portfolio to EQB's platform under OSFI and Ministerial oversight. All analysis is ex-ante scored as forward-looking predictions from the announcement date grounded in EQB press releases, OSFI B-13 guidelines, PIPEDA, and industry reporting. The assignment produced a 40-risk register, a risk calculation sheet, and dual heat map visualizations (Excel and Python), described below.

Risk Scoring Methodology

Each risk was scored on the course-specified non-standard scale. Likelihood: 1 (Very Unlikely), 3 (Somewhat Unlikely), 5 (50-50), 7 (Somewhat Likely), 9 (Very Likely). Impact: 1 (Very Low), 2 (Somewhat Low), 4 (Moderate), 6 (Somewhat High), 8 (High), 9 (Extremely High). Risk Score = Likelihood × Impact, ranging from 1 to 81. Three thresholds were applied across all deliverables: High (≥ 45), Medium (21–44), and Low (≤ 20). The 40 risks span eight categories and include three positive risks (opportunities) scored identically and flagged with ★ in all outputs.

The distribution across the 40 risks: 7 High (including R-01 Cybersecurity breach, Score 81; R-38 Cross-sell opportunity, Score 81), 24 Medium, and 9 Low. External research including OSFI regulatory updates, Canadian banking sector reports, and competitor analysis of TD, RBC, Koho, and Neo Financial informed scoring decisions.

Risk Heat Map (Excel Manual)

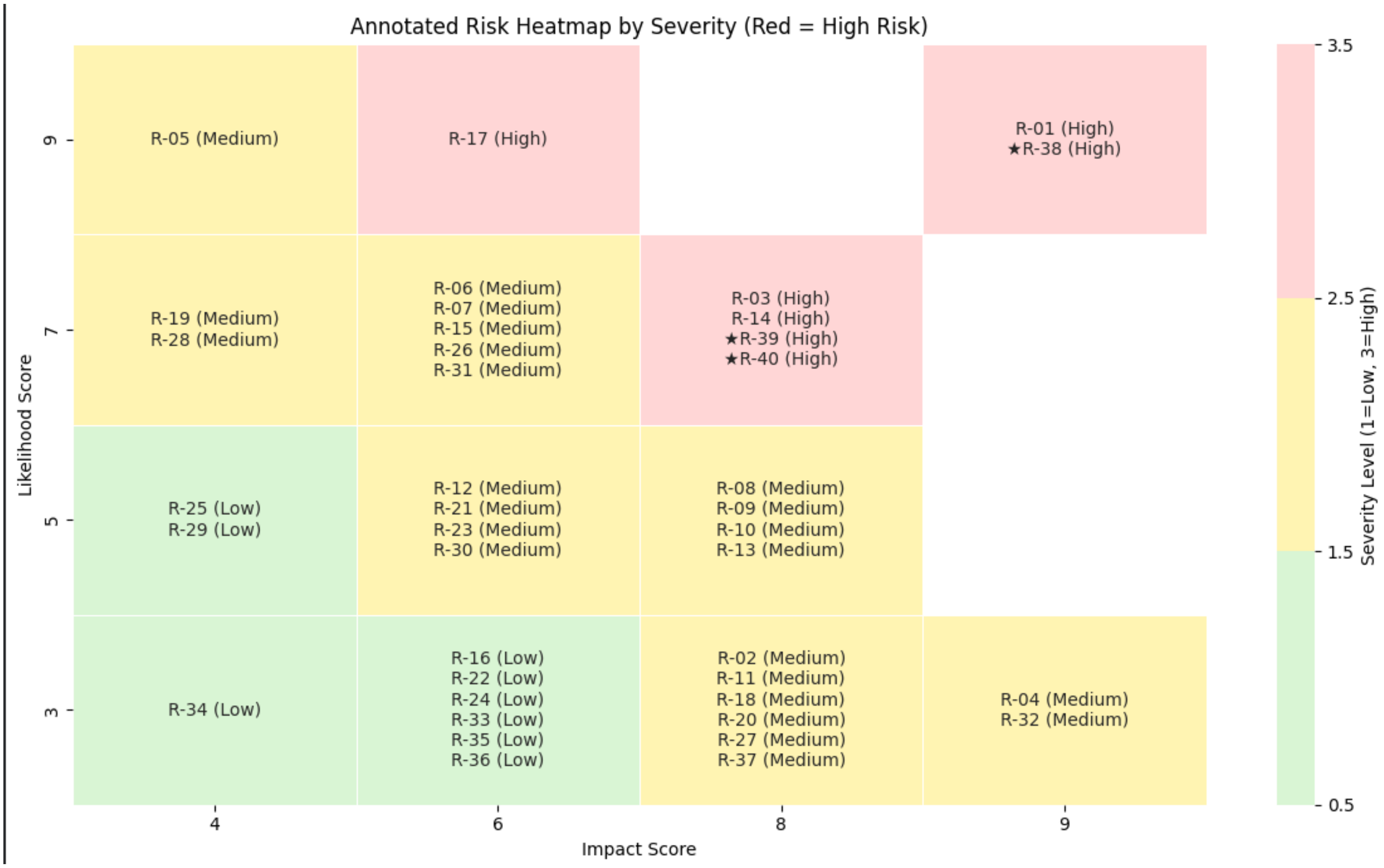
The Excel heat map (Heat Map tab) is a 5×6 grid with Likelihood on the Y-axis (9 to 1, descending) and Impact on the X-axis (1 to 9, ascending). Each cell displays the computed Risk Score and the IDs of all risks at that coordinate, with ★ prefixes on positive risks. Background fill red (High ≥ 45), yellow (Medium 21–44), green (Low ≤ 20) is applied per cell based on the zone of the cell's maximum score. The heat map is contained in the same workbook as the scoring table so that values are traceable directly from calculation to visualization.

Scoring Scale: Likelihood (1/3/5/7/9) × Impact (1/2/4/6/8/9) | High ≥ 45 | Medium 21–44 | Low ≤ 20 | ★ = Positive Risk

Likelihood ↓ / Impact →	IMPACT SCORE →					
LIKELIHOOD SCORE →	1 Very Low	2 Somewhat Low	4 Moderate	6 Somewhat High	8 High	9 Extremely High
9 Very Likely	Score: 9	Score: 18	Score: 36 R-05	Score: 54 R-17	Score: 72	Score: 81 R-01 ★R-38
7 Somewhat Likely	Score: 7	Score: 14	Score: 28 R-19 R-28	Score: 42 R-06 R-07 R-15 R-26 R-31	Score: 56 R-03 R-14 ★R-39 ★R-40	Score: 63
5 50-50	Score: 5	Score: 10	Score: 20 R-25 R-29	Score: 30 R-12 R-21 R-23 R-30	Score: 40 R-08 R-09 R-10 R-13	Score: 45
3 Somewhat Unlikely	Score: 3	Score: 6	Score: 12 R-34	Score: 18 R-16 R-22 R-24 R-33 R-35 R-36	Score: 24 R-02 R-11 R-18 R-20 R-27 R-37	Score: 27 R-04 R-32
1 Very Unlikely	Score: 1	Score: 2	Score: 4	Score: 6	Score: 8	Score: 9

Risk Heat Map (Python Notebook)

The Jupyter notebook loads the risk register Excel file directly via pandas (pd.read_excel), derives severity labels using the same classify_severity() threshold logic, and produces an annotated seaborn heat map using a pivot table (groupby Likelihood × Impact). A discrete ListedColormap replicates the red/yellow/green zones. The notebook also runs a Random Forest classifier (100 trees, 5-fold CV) achieving 93% mean accuracy correctly classifying all 24 Medium risks, 6 of 7 High, and 7 of 9 Low risks. Libraries used: pandas, numpy, matplotlib, seaborn, scikit-learn (RandomForestClassifier, cross_val_score, cross_val_predict).



Observations

- **Highest priority:** R-01 (Cybersecurity breach, Score 81) and R-38 (Cross-sell opportunity, Score 81) share the top score the acquisition's greatest threat and greatest upside stem from the same integration.
- **High-risk cluster:** Seven High risks concentrate in the top-right heat map zone (Likelihood 7–9 × Impact 8–9), indicating the integration execution phase carries the most concentrated exposure.
- **Positive risks are High-zone:** All three opportunities (R-38, R-39, R-40) score in the High zone, confirming the acquisition's strategic rationale is strong but contingent on mitigating the surrounding operational and regulatory risks.
- **Medium risks dominate:** 24 of 40 risks are Medium, spanning financial, operational, and regulatory categories requiring sustained management attention rather than isolated crisis response.

Conclusion

The dual Excel-and-Python approach to heat map development demonstrates that risk visualization can be manual, reproducible, and analytically rigorous simultaneously. The 40-risk register, grounded in verifiable external sources and the course scoring scale, gives EQB management a transparent, threshold-consistent view of where integration risks concentrate and where strategic opportunities can be unlocked. The Python implementation adds the capability to re-run the entire analysis automatically as the risk register evolves.

References

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